Research Training Group 2721

From Hydrogen to Helium Isotope Separation

Elvira Gouatieu Dongmo

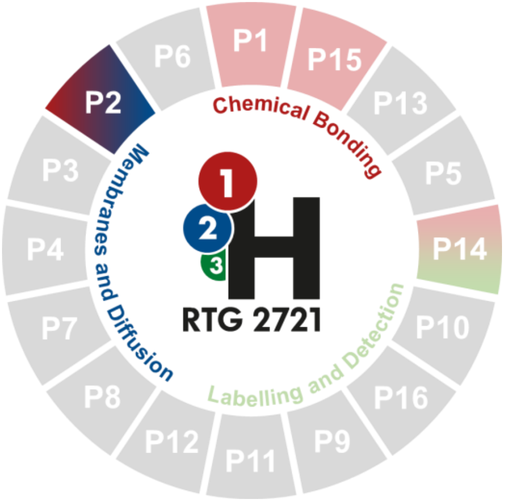
TU-Dresden & Helmholtz-Zentrum Dresden-Rossendorf (HZDR)

Funded by

DFG Deutsche Forschungsgemeinschaft

German Research Foundation

P2: From Hydrogen to Helium Isotope Separation

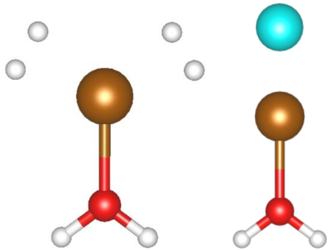


Overarching Research Question

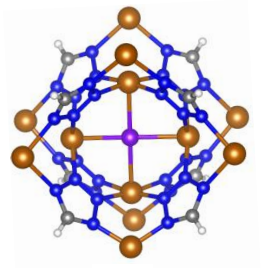
Can we control the activity of **adsorption sites** to achieve high isotope selectivity?

Research Hypothesis

- **Gas phase clusters** involving undercoordinated Cu(I) sites are suitable models for materials hosting undercoordinated Cu(I) such as **zeolites** and **MOFs**



Strong adsorption site



MOFs: Metal-Organic Frameworks

S. Haque
P1/1

F. Kreuter
P14/a

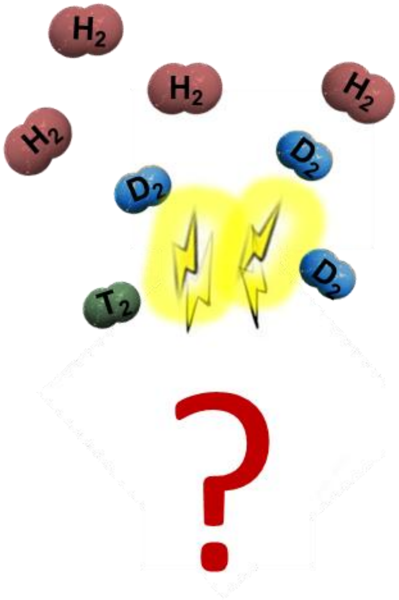
R. Barrett
P15/2

Publications

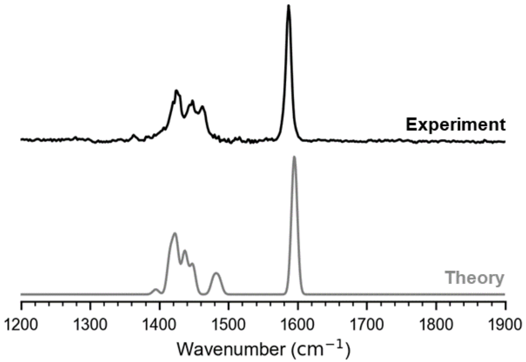
E. G. Dongmo, **S. Haque**, **F. Kreuter**, **T. Wulf**, **J. Jin**, R. Tonner-Zech, T. Heine, K.R. Asmis, *Chem. Sci.* **2024**, *15*, 14635
E. G. Dongmo, **T. Wulf**, S. Das, T. Heine, submitted

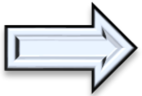
Approach

Gas Phase Model Systems Characterized via IRPD

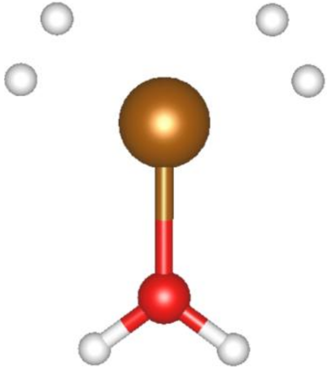


IRPD spectra





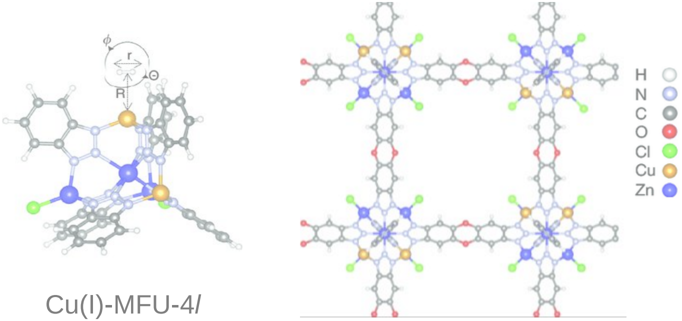
Enhanced Isotope selectivity



D_2/H_2 selectivity

Cu^+—H_2 Interaction

Very Strong Attraction to H_2 by Cu(I)

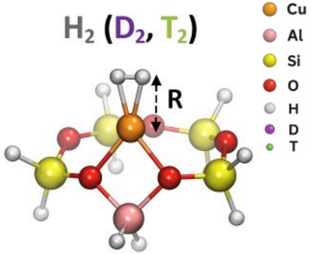


Cu(I)-MFU-4*l*

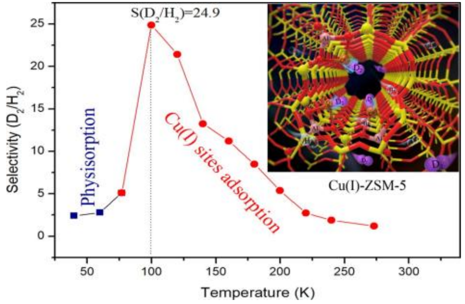
☞ $\Delta_{ad}H(H_2)$ = – 32.7 kJ·mol^{-1} and α(D_2/H_2) =**11 (T = 100 K)**

☞ Predicted α(T_2/D_2) ≈ **3.7 (T = 100 K)**

Efficient D_2/H_2 and T_2/D_2 Separation on Cu(I) Zeolites



Cu(I)-ZSM-5



☞ α(D_2/H_2) = **24.9 (T = 100 K)**

☞ Predicted α(T_2/D_2) ≈ **3.5 (T = 100 K)**

In real materials, many adsorption sites are probed simultaneously

T. Heine

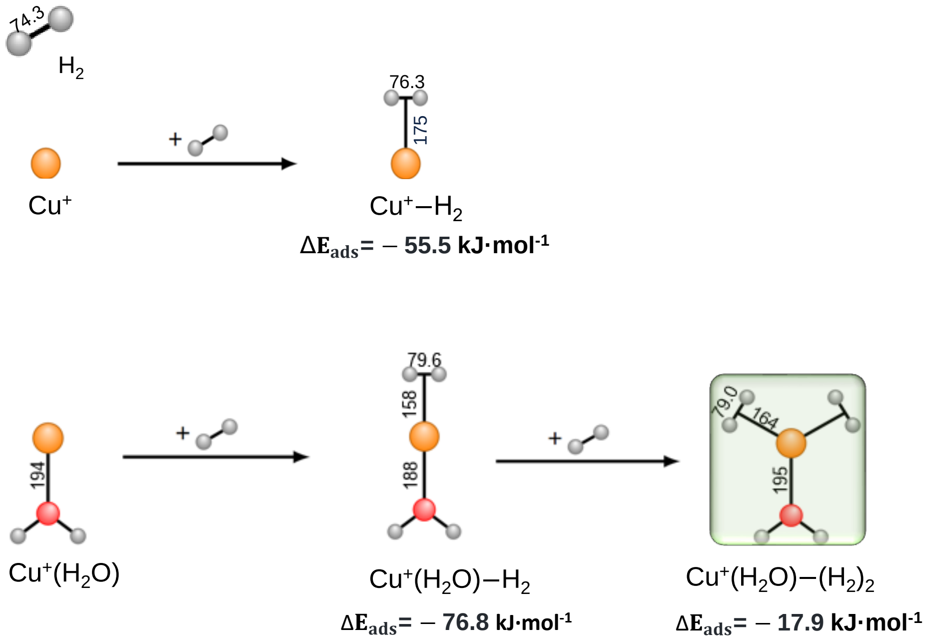
M. Hirscher

M. Hirscher

I. Weinrauch, I. Savchenko, D. Denysenko, S.M. Souliou, H. H. Kim, M. Le Tacon, …, **M. Hirscher** & T. Heine, *Nat. Commun*. **2017**, *8*, 14496

R. Xiong , L. Zhang , P. Li, W. Luo, T. Tang, B. Ao, G. Sang, C. Chen, X. Yan, J. Chen, **M. Hirscher**, *Chem. Eng. Journal.* **2020**, *391*, 123485

H_2 Interaction with Bare Cu^+ and $Cu^+(H_2O)$



H Cu O

All bond lengths are given in picometers (pm)

Energy values are ZPE-corrected

- The H-H bond is weakened upon interaction with Cu^+
- Stronger binding in $Cu^+(H_2O)$-H_2 compared to Cu^+-H_2

Optimizations and frequencies for H_2 isotopologues → **CCSD(T)/aug-cc-pVTZ-(PP)**

Research stay at Uni Graz

D. Boese

J. Hoja

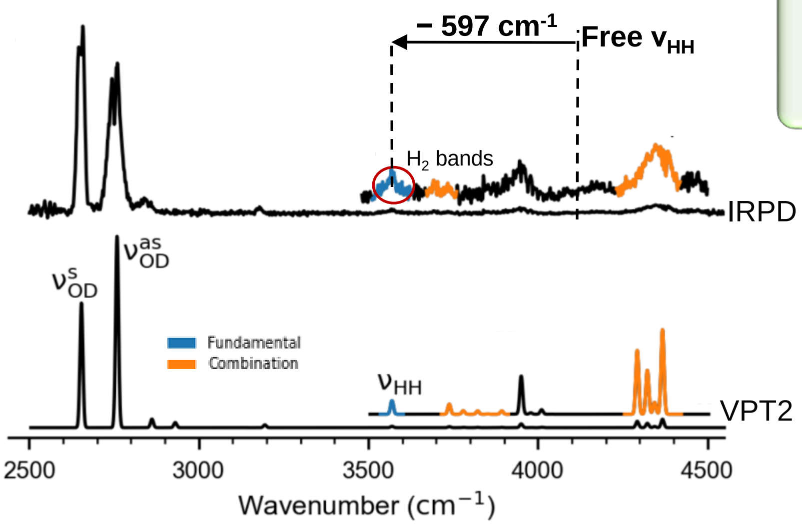
VPT2 : 2nd order Vibrational Perturbation Theory

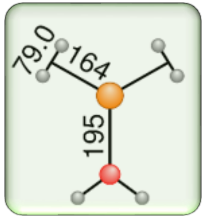
CFOUR : Coupled-Cluster techniques for Computational Chemistry

E. G. Dongmo, **S. Haque**, **F. Kreuter**, **T. Wulf**, **J. Jin**, R. Tonner-Zech, T. Heine, K.R. Asmis, *Chem. Sci.* **2024**, *15*, 14635

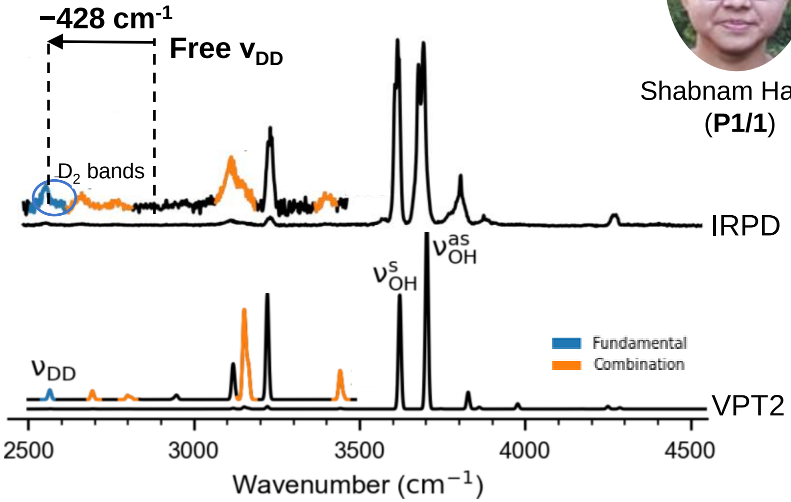
Vibrational spectra: IPRD vs. VPT2

$Cu^+(D_2O)(\mathbf{H_2})_2$





$Cu^+(H_2O)(\mathbf{D_2})_2$



Shabnam Haque
(**P1/1**)

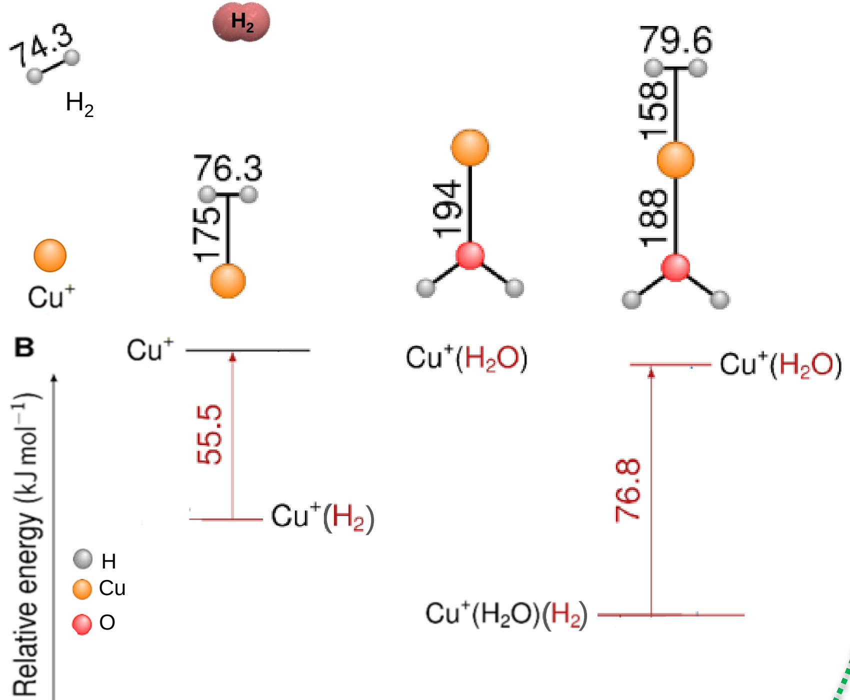
- IRPD spectra match the calculated frequencies using **VPT2**.

- Large $\mathbf{\nu_{HH}}$($\mathbf{\nu_{DD}}$) red-shift : **− 597 cm^{-1}** (**− 428 cm^{-1}**).
- Elongation of the H-H distance by **4.7 pm**

E. G. Dongmo, **S. Haque**, **F. Kreuter**, **T. Wulf**, **J. Jin**, R. Tonner-Zech, T. Heine, K.R. Asmis, *Chem. Sci.* **2024**, *15*, 14635

IRPD : Infrared photodissociation spectroscopy

ZPE and Separation of H_2, HD and D_2 on Cu(I) Complexes



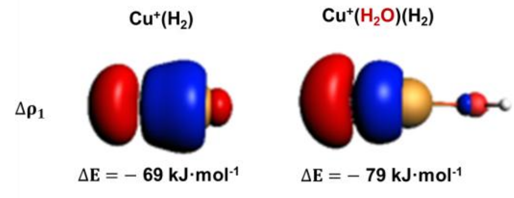
H$_2$ binds even **more strongly** upon addition of a single **H_2O ligand**

Adding a single water Molecule

Increase in the electrostatic attraction ($\Delta\Delta E_{elstat}$ **= − 17 kJ·mol^{-1}**)

Florian Kreuter
(**P14/a**)

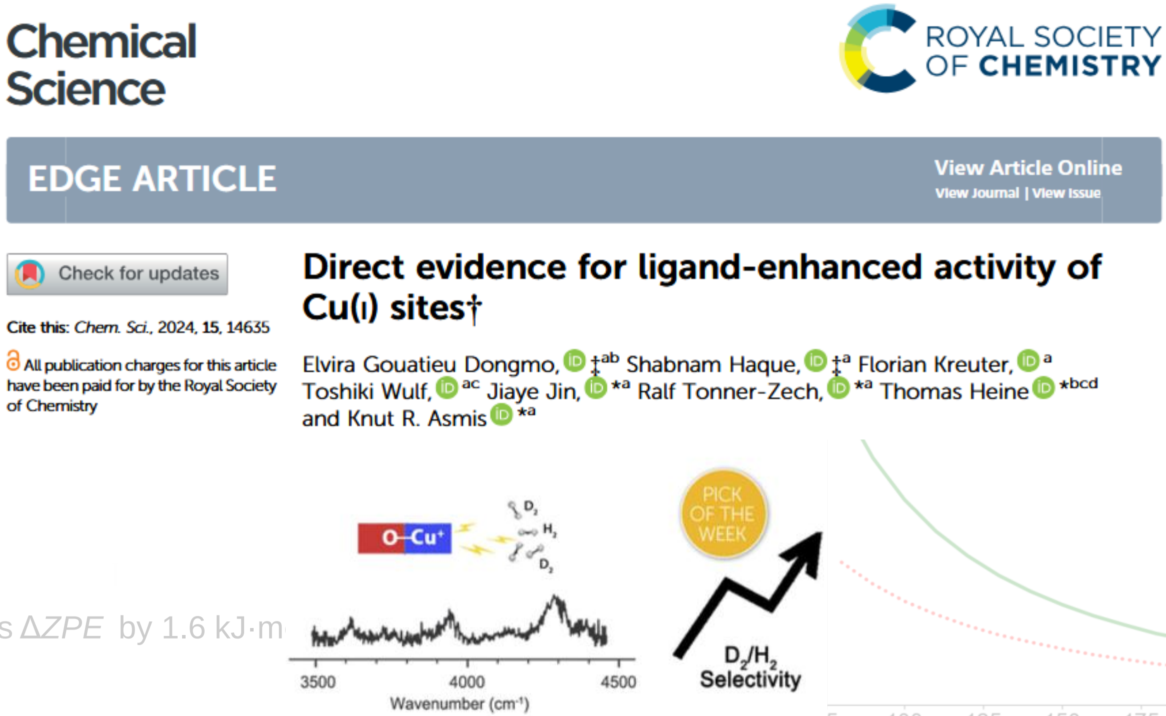
σ-donation increases by +**10 kJ·mol^{-1}**



Lower Pauli repulsion, allows H_2 to bind strongly

E. G. Dongmo, **S. Haque**, **F. Kreuter**, **T. Wulf**, **J. Jin**, R. Tonner-Zech, T. Heine, K.R. Asmis, *Chem. Sci.* **2024**, *15*, 14635

ZPE and DHI Separation on Cu(I) Complexes



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Cite this: *Chem. Sci.*, 2024, 15, 14635

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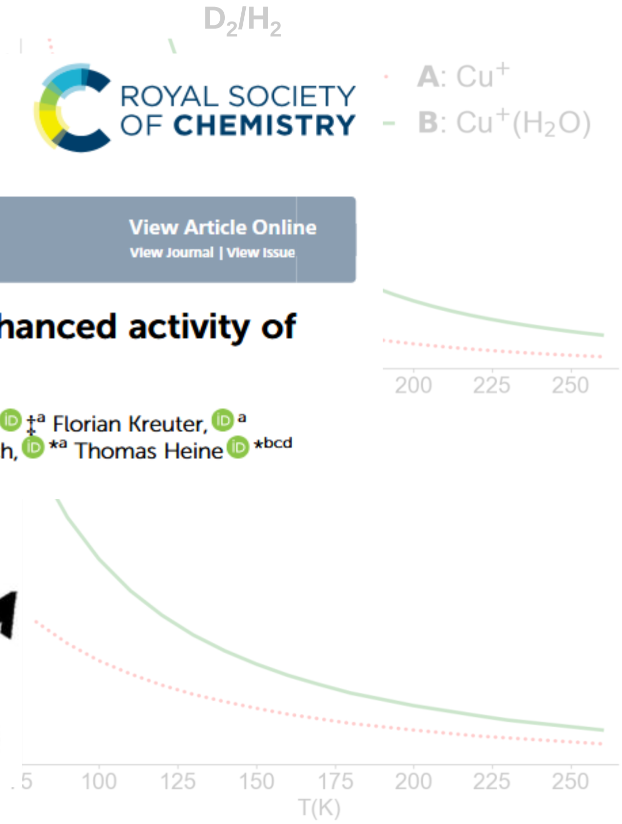
Direct evidence for ligand-enhanced activity of Cu(I) sites†

Elvira Gouatieu Dongmo, ‡[ab] Shabnam Haque, ‡[a] Florian Kreuter, [a] Toshiki Wulf, [ac] Jiaye Jin, *[a] Ralf Tonner-Zech, *[a] Thomas Heine *[bcd] and Knut R. Asmis *[a]



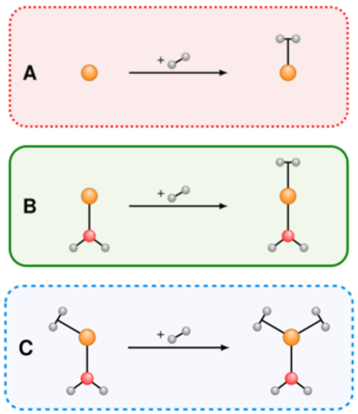
H_2O ligand increases ΔZPE by 1.6 kJ·m

Zero-point energy (ZPE) is obtained through nuclear quantum effects (NQEs)

$\Delta ZPE = ZPE(D_2) – ZPE(H_2)$

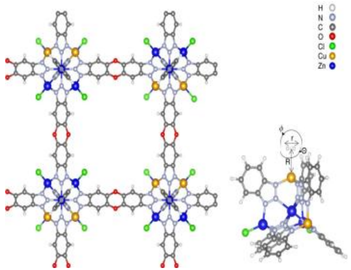
$Cu^+(H_2O)$ shows higher separation factor than bare Cu^+

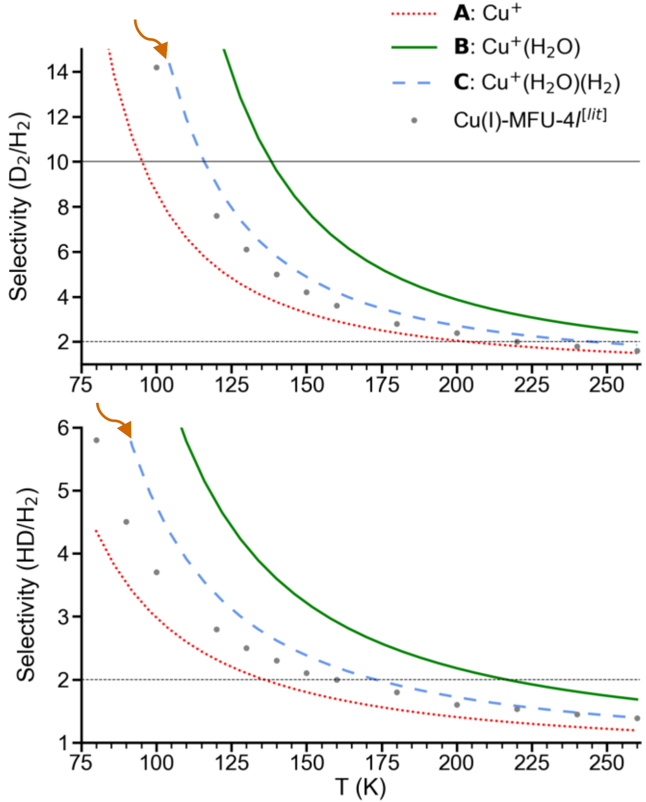
D_2/H_2 and HD/H_2 Selectivity: Comparison with Cu(I)-MFU-4*l*



Benchmark:

Cu(I)-MFU-4*l*





$\alpha_{D_2/H_2} \approx$ 2 at T= 255 K

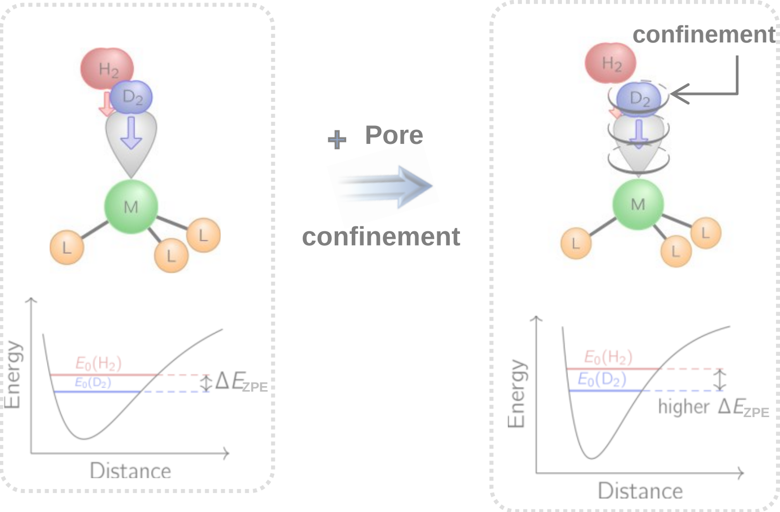
$\alpha_{D_2/H_2} \approx$ 10 at T= 150 K

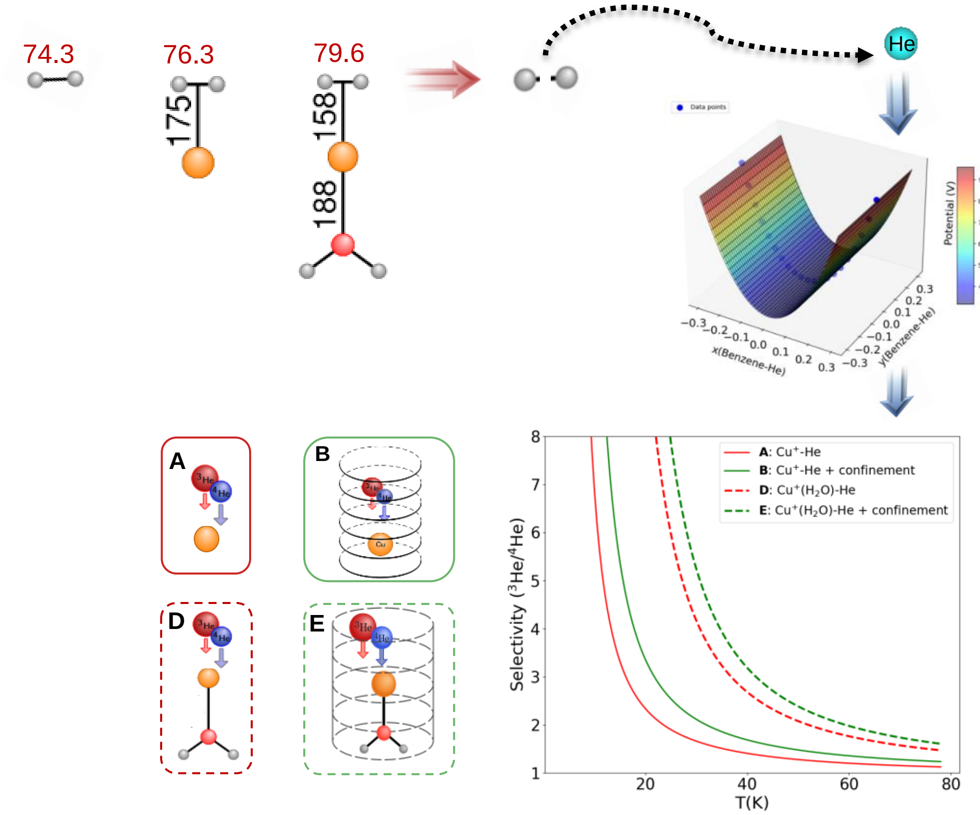
How to Improve this Separation factor?

I. Weinrauch, I. Savchenko, D. Denysenko, S.M. Souliou, H. H. Kim, M. Le Tacon, …, **M. Hirscher** & T.Heine, *Nat. Commun*. **2017**, *8*, 144960

Thermodynamic Limits in a Numerical Model

Can we use confinement inside the pore?





Model: spatially well-tuned environment

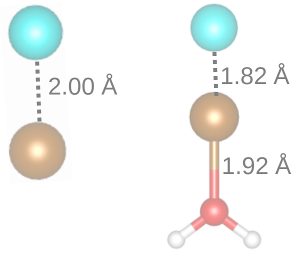
$\Delta ZPE = ZPE(D_2) - ZPE(H_2)$

What enhancement is possible ?

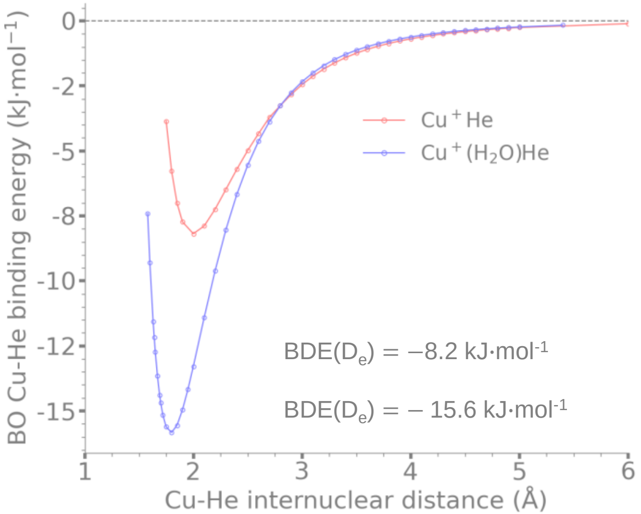
At what temperature is separation possible? 77 K?

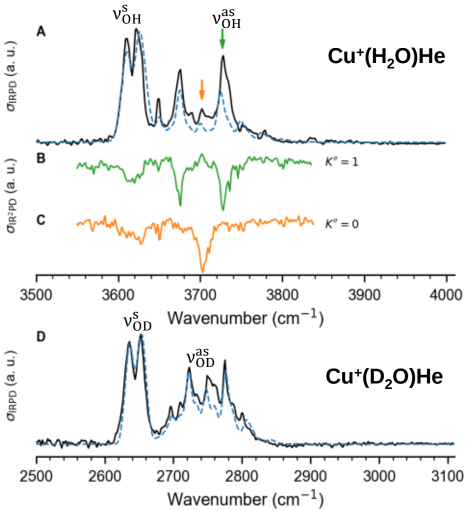
E. G. Dongmo, **T. Wulf**, T. Heine, in preparation

$Cu^+(H_2O)$–He and $Cu^+(D_2O)$–He



H_2O ligand enhance the Cu^+–He bond strength as in H_2 case





Shabnam Haque
(P1/1)

$Cu^+(H_2O)He$		
Exp.	VPT2	Ass.
3609	3608 (120)	ν^s_{OH}
3688	3688 (285)	ν^{as}_{OH}

$Cu^+(D_2O)He$		
Exp.	VPT2	Ass.
2653	2644 (74)	ν^s_{OD}
2749	2748 (142)	ν^{as}_{OD}

Geometry: CCSD(T)/aug-cc-pVTZ
Electronic energy: CCSD(T)/aug-cc-pVTZ
Frequencies: VPT2/MP2/def2-TZVPP

S. Haque, E. G. Dongmo, **T. Wulf**, **J. Jin**, T. Heine, K.R. Asmis, in preparation

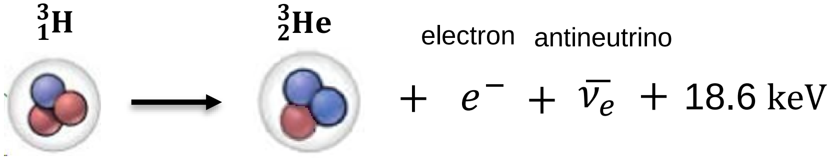
Why Helium?

New Way to Achieve Nuclear Fusion!

E. Mazzucato, *Fundam. Plasma Phys*. **2023**, *6*, 100022

Helium-3 (^3He)

- ^3He is **used for cryogenic applications**
- ^3He production still mainly rely on **tritium decay**



$$^{3}_{1}\mathrm{H} \longrightarrow {}^{3}_{2}\mathrm{He} + e^{-} + \bar{\nu}_e + 18.6\ \mathrm{keV}$$

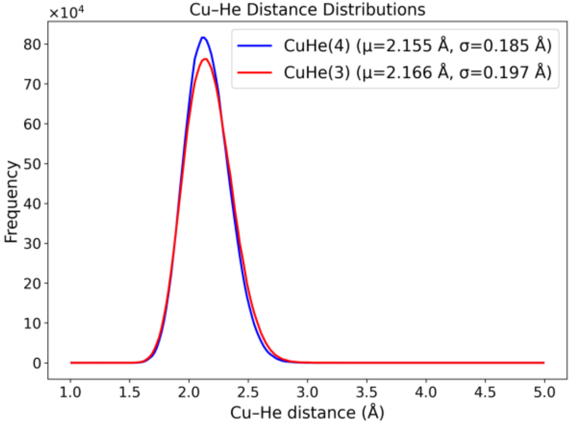
M. Shimada, B.J. Merrill, *Nucl. Mater. Energy*. **2017**, *12*, 699

This calls for alternative sources!

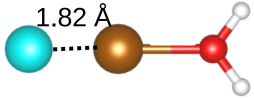
W. Kempiński, Sz. Łoś, Z. Trybuła, M. Chorowski, J. Poliński, J. Niechciał, T. Jaskólski, K. Chołast, A. Kociemba, *Sep. Purif. Technol.* **2019**, *210*, 276

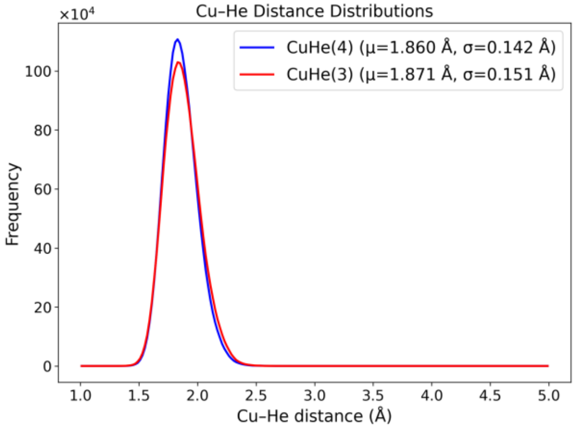
Machine Learning for Approximating NQEs

- The difference between Cu-^{3}He and Cu-^{4}He bond distance increases in presence of H_2O ligand



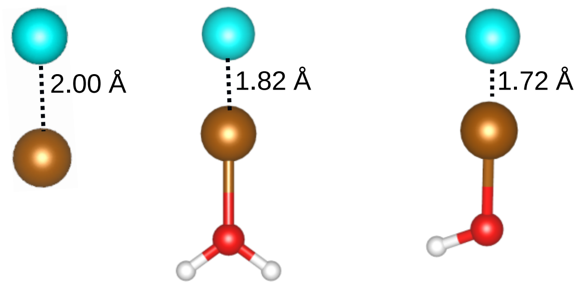
Rhyan Barrett
(**P15/2**)





- Ring polymer dynamics simulations utilized 128 beads
- Simulation time of 1 GPUh vs nearly 150,000 CPUh for Cu-He
- Simulation time of 2 GPUh vs nearly 215,000 CPUh for H_2O-Cu-He

Small Cu(I) Complexes

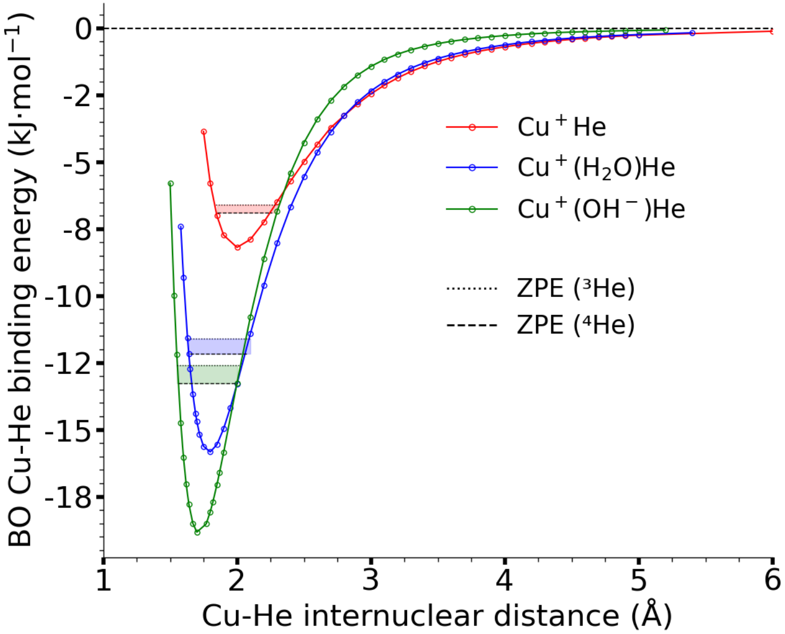


- ΔZPE increases with increasing adsorption energy
- Strongest adsorption → neutral complex: $Cu^+(OH^-)He$
- Upon anharmonic correction with VPT2

$Cu^+(OH^-)He$ ΔZPE → 0.7 kJ·mol^{-1}

Cu^+He ΔZPE → 0.4 kJ·mol^{-1}

ΔZPE = ZPE(^{4}He) – ZPE(^{3}He)



Geometry: CCSD(T)/aug-cc-pVTZ

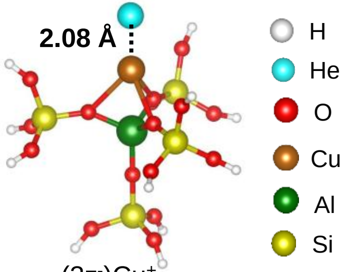
Zero point energy: DLPNO-CCSD(T)@TightPNO/cc-pVTZ (aug-cc-pVTZ for Cu and He) (numerical method)

E. G. Dongmo, **T. Wulf**, S. Das, T. Heine, submitted

Materials Models

S.Das
Marie S. Curie Fellow
and Postdoc at TUD

Zeolites

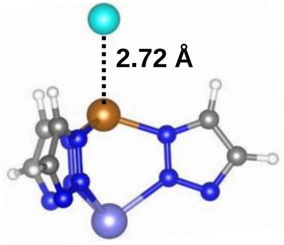


(3z$^-$)Cu$^+$

$E_{ads} = -3.5$ **kJ·mol^{-1}**

α =2.3

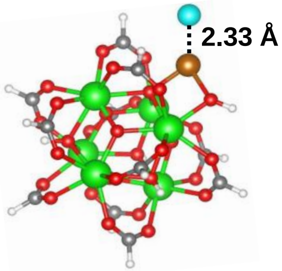
MOFs



Cu(I)-MFU-4*l*

$E_{ads} = -.0.9$ **kJ·mol^{-1}**

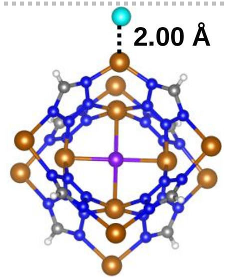
α =1.3



Cu(I)/UiO-66

$E_{ads} = -2.3$ **kJ·mol^{-1}**

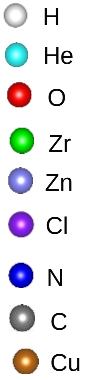
α =2.4



Cu[(Cu$_4$Cl)(ttpm)$_2$]$_2$

$E_{ads} = -4.1$ **kJ·mol^{-1}**

α =3.9



$\alpha = \alpha(^3\text{He}/^4\text{He})$: isotope selectivity at T = 20 K
energy values are ZPE-corrected

Geometry and Electronic energy: DLPNO-CCSD(T)@TightPNO/cc-pVTZ (aug-cc-pVTZ for Cu and He)

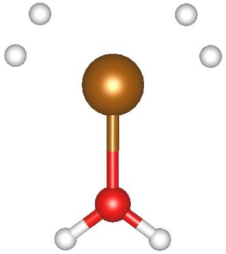
- Cu(I) complexes are adsorbing strongest if coordinate to two or three ligands
- **Cu(I) anchored at tetrahedral sites in Cu[(Cu$_4$Cl)(ttpm)$_2$]$_2$**

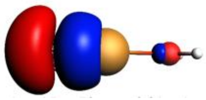
ttpm = Tetrakis(4-tetrazolylphenyl)methane

E. G. Dongmo, **T. Wulf**, S. Das, T. Heine, submitted

Summary and Outlook

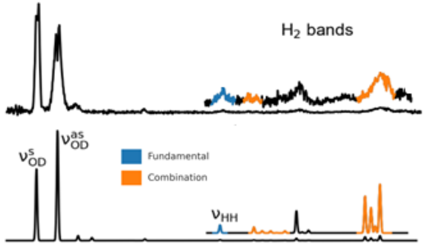
Strong Adsorption Site





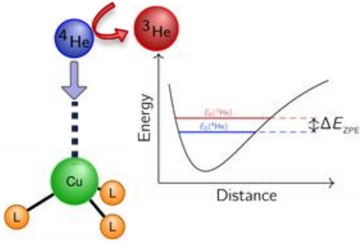
Origin of strong bonding

Vibrational Spectra

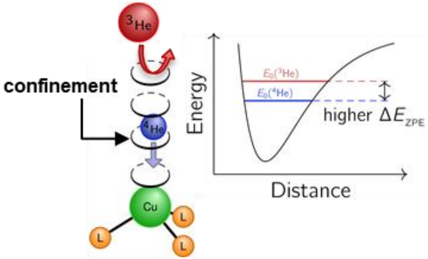


E. G. Dongmo, **S. Haque**, **F. Kreuter**, **T. Wulf**, **J. Jin**, R. Tonner-Zech, T. Heine, K.R. Asmis, *Chem. Sci.* **2024**, *15*, 14635

^{3}He/^{4}He Selectivity Enhanced



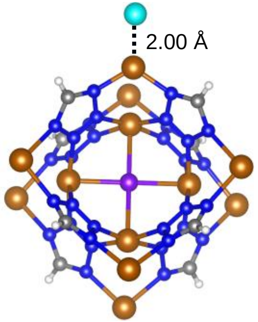
attractive surface adsorption site



optimal-sized pore

E. G. Dongmo, **T. Wulf**, T. Heine, in preparation

Cu[(Cu$_4$Cl)(ttpm)$_2$]$_2$



α(^{3}He/ ^{4}He) ≈ 3.9 at T= 20 K

α(^{3}He/ ^{4}He) ≈ 1.3 at T= 77 K

E. G. Dongmo, **T. Wulf**, S. Das, T. Heine, submitted

MOFs and MOF-Glasses for Helium Isotope Separation

Hoi Ri Moon
^{123}H Mercator Fellow

S. Das
Marie S. Curie Fellow
and Postdoc at TUD

Backup slides

Detailed Calculations of Adsorbed He Isotope Selectivity

$\boldsymbol{\alpha(^{3}He/^{4}He) = K_{ad}(^{3}He)/K_{ad}(^{4}He)}$

where $\boldsymbol{K_{ad}(^{3}He)}$ and $\boldsymbol{K_{ad}(^{4}He)}$ represent the **equilibrium adsorption constants** for ^{3}He and ^{4}He, respectively, given by

$$K_{ad}(^{3}He) = \frac{q^{ad}_{^{3}He}}{q^{g}_{^{3}He}} \text{ and } K_{ad}(^{4}He) = \frac{q^{ad}_{^{4}He}}{q^{g}_{^{4}He}}$$

with the partition functions

$$q^{ad}_{^{3}He} = q^{ad,vib}_{^{3}He}(\nu_{M-^{3}He}) \text{ and } q^{ad}_{^{4}He} = q^{ad,vib}_{^{4}He}(\nu_{M-^{4}He}).$$

Here M denotes the metal adsorption site, that is, Cu(I) in this study.

$$q_i^{ad,vib} = \sum_{\nu=0}^{\infty} e^{-\frac{E_i(\nu)}{R\cdot T}}$$

where R is the ideal gas constant, and T is the temperature.